A Low Cost Edge Computing and LoRaWAN Real Time Video Analytics for Road Traffic Monitoring

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***Abstract*—Traffic congestion is a major problem in many cities. It happens due to the demand-supply imbalance in the transportation network and poor management. Traffic flow slows down when the number of vehicles that travels on the road increases or the roadway capacity decreases due to various reasons. In order to solve this problem, different solutions are proposed to provide reliable, real-time transport management services in an Intelligent Transportation System (ITS). In this paper, we propose a novel real-time video analytics using low-cost IoT devices and LoRaWAN networks to realize new services and applications that include traffic management through IoT edge computing. The use of LoRaWAN for such application is our main contribution. We retrain YOLO v3 object detection machine learning model (transfer learning) for vehicle detection and counting, to make it lightweight and fast enough to be able to run on a Raspberry Pi, a single-board computer with limited RAM. The edge node, with low-cost smart camera and connectivity through LoRaWAN networks counts the number of vehicles using real-time video analytic and report only traffic count to the server. This experimental work provides insight into the applicability of a low-cost IoT system to traffic management with a resource-constrained environment. Real-world video analysis of vehicle detection and counting show the effectiveness of the designed solution. The results demonstrate the effectiveness of the proposed approach.**

I. Introduction

Many countries, especially developing ones, suffer from traffic congestion, poorly planned road networks, and lack of access to public transportation facilities. This results in increases in fuel consumption, pollution level, monetary losses, massive delays, and decreased productivity [1]. Road traffic is a complex phenomenon, where various entities (pedestrians, cars, trucks, buses, bicycles, etc.) interact over the shared infrastructure. Traffic flow prediction is a challenging problem in transportation planning and car navigation systems. Traditionally, traffic flows used to be estimated using historical data. However, traffic flow prediction cannot be done relying solely on past traffic data due to events which cannot be predicted a priory, like accidents, road closure, or weather conditions.

Due to infrastructure constraints and rising number of vehicles, traffic management and control is a complex task and requires the application of dedicated intelligent transport systems integrating sensing, communication and advanced algorithms to collect, communicate and perform real time analytics on traffic data. The information about the number of vehicles and their types is helpful in reducing travel times and emissions. Precise traffic data allows us not only to increase effectiveness of traffic control, but also to adapt management policy to changing conditions and predict infrastructure bottlenecks [2]. The traffic data can be provided by traffic monitoring systems that are usually integrated with road infrastructure using roadside sensors to collect various types of traffic data, including flows, density, occupancy, velocity, and travel time. Such systems allow detecting and classifying the vehicles in selected areas by using data from sensors (inductive loops, video-detectors, magnetometers, etc.). A major drawback of these solutions integrated with the infrastructure is low flexibility and significant maintenance cost. To overcome these drawbacks, applications of new technologies in traffic monitoring are considered which often entail high installation and configuration costs.

Nowadays,with advancement of the Internet of Things (IoT), wireless communication and Artificial Intelligence (AI), ITS is further improved by using intelligent sensors, wireless communication technology, distributed data, information processing, and Internet technology to achieve the efficient exchange of information and sharing between human and vehicle, vehicle and vehicle, and vehicle and roadside infrastructure.

In this paper, we propose a vehicle traffic monitoring system based on Edge Computing [3] and LoRaWAN [4] using low cost camera sensors, a Raspberry PI IoT platform and machine learning video analytics to count the number of vehicles in real time for traffic monitoring and reporting.

There are many well known approaches for object detection like Single Shot Detector (SSD) [5], You Only Look Once (YOLO)v3 [6], and Faster-RCNN [7]. The choice of a right object detection method is critical and depends on the problem to solve as well as on the speed and accuracy requirements.

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Although Faster-RCNN achieves excellent object detection accuracy, it is computationally intensive and not suitable for use in real time applications to detect objects like pedestrians, vehicles, etc [8] [9] [10]. Since speed is the main requirement for real-time vehicle detection and counting, and given that YOLO v3 is faster than SSD and Faster RCNN [11], we chose YOLO v3 as our object detection machine learning model (transfer learning) for vehicle detection and counting. Moreover, it is lightweight so it can run on a single-board computer with limited RAM.

Vehicular fog computing, an integration of fog computing and vehicular networks, enables real-time and location-aware network responses for better distributed traffic management. Unlike the traditional IoT architectures where smart devices are connected to cloud computing centers to transfer and process data, we propose to leverage the advantage of fog computing to process locally instead of transferring huge amounts of data from geographically distributed devices which create network overload and latency, especially critical in developing countries and delay-sensitive applications.

Fog computing augments cloud computing for near real-time data analytics, faster localized actions, and efficient data management and operations [12]. However, implementing fog computing is more challenging due to the distributed nature of fog systems [13]. Authors in [14] propose a context aware fog computing framework for intelligent transportation systems (ITS) with multiple tiers: IoT tier, fog service tier, and global cloud service tier.

Similar works [15] showed the applicability of machine learning based traffic monitoring using high end embedded platforms (Jetson TX2), however, it cost in the order of 1000 USD and lacked communication options. A paper [16] also proposes urban traffic analysis using convolutional neural networks and closed-circuit television (CCTV) datasets for detection and classification of road users, but is based on high-end devices and therefore more expensive.

The novel aspects of our study include:

- Proposal of a low cost and real-time vehicle detection and counting system, which is suitable for crowdsensing applications;
- Field experiment and verification of the LoRaWAN based traffic monitoring system in real-traffic conditions.

The article is organized as follows: Section 2 presents related works. Section 3 describes the proposed solution. Section 4 describes the experiment setup and results reached. Section 5 concludes the paper with comments and final remarks.

II. Related Works

There have been many studies in the literature on collecting traffic data for road traffic monitoring including inductive loop detectors [17], RADAR [18], etc. The advent of IoT and wireless communication has enabled the development of advanced traffic monitoring technologies relaying on various smart sensors: Papers based on inputs from (i) static roadside sensors like cameras [19] [20] [21], microphones [22], radio frequency, and Bluetooth beacon signal strength [23] [24] or (ii) from probe sensors on vehicles.

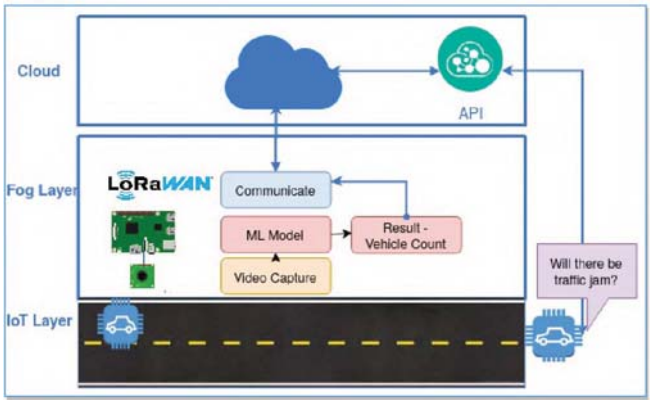


Fig. 1. System Architecture

Recently, some works employing IoT, edge computing and machine learning for vehicle classification and counting have emerged. Kerouh et. al [19] developed an Android application to estimate traffic density in real time using Smartphone's camera and real-time video processing. The system autonomously computes the vehicle flow density in real time and saves all the relevant information in the smartphone's memory. The authors claim that they validated the app under real conditions in different meteorological conditions and varying intrinsic and extrinsic camera parameters and report the effectiveness and the applicability of the proposed application. There are also some recent works using Convolutional Neural Network (CNN) based object detection model for vehicle classification and counting, using IoT devices and validated on New Delhi roads for traffic monitoring [20]. A paper [15] also demonstrated the applicability and efficiency of real-time traffic monitoring using IoT devices. Similar works [15] showed the applicability of machine learning based traffic monitoring using high end embedded platforms (Jetson TX2), which cost in the order of 1000 USD. A paper [16] proposes urban traffic analysis using convolutional neural networks and closed-circuit television (CCTV) datasets for detection and classification of road users. A study [25] also demonstrated the performance of vision-based traffic flow data analysis algorithm using a hardware device, Intel Viewmont video analytics coprocessor. Similar work by Aneiba et. al. [26] used inductive loop technology with LoRaWAN for road traffic data monitoring.

To the best of our knowledge, there are limited few previous research work on employing low cost technologies such as LoRaWAN, edge computing, video analytics and models for intelligent transport systems, especially for traffic flow prediction. Thus, our solution provides edge analytics for near-real time traffic data collection and decision in ITS offering actionable information back to the users to help reducing traffic congestion.

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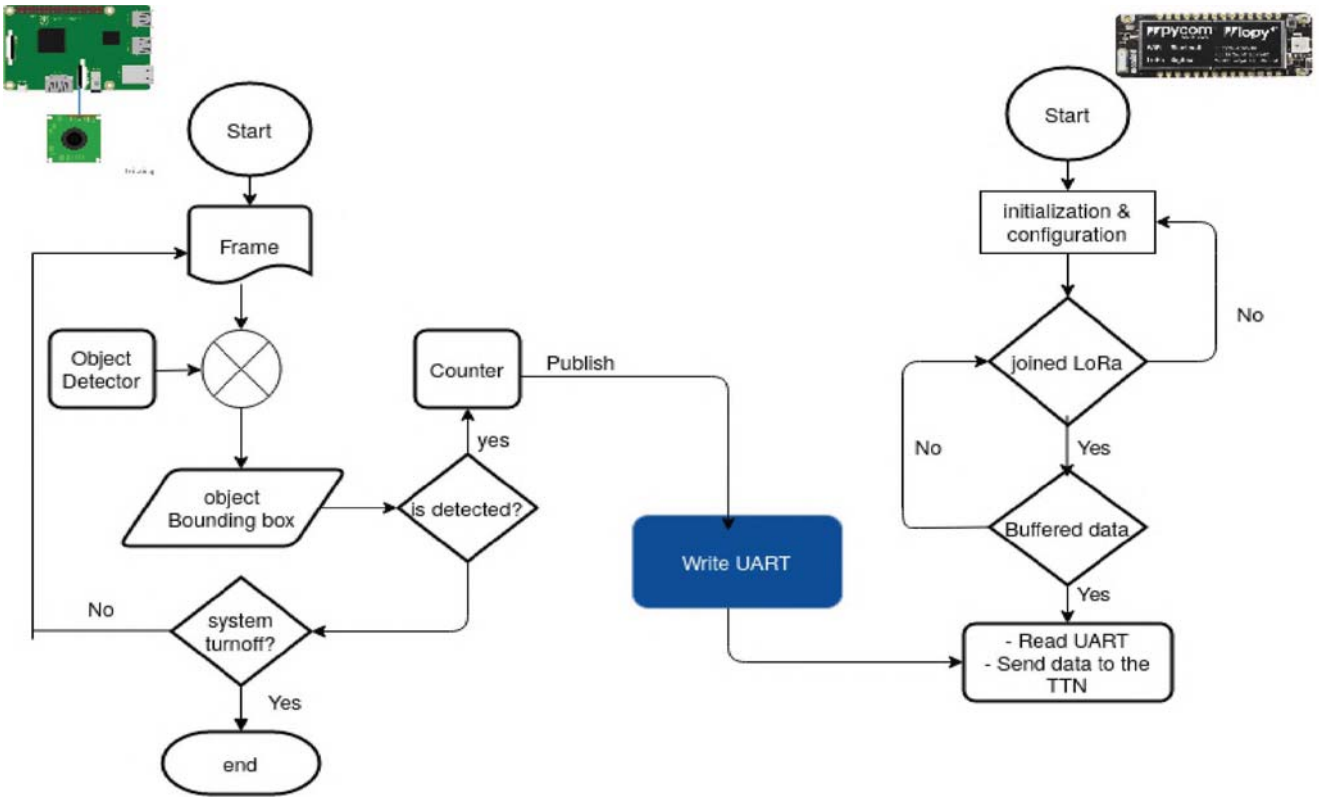


Fig. 2. Flow chart of the steps followed by the system.

III. Proposed System

In this section we first introduce the operational requirements of the system. We then describe the proposed architecture along with different hardware, software, and communication solutions including machine learning models which enable vehicle detection and counting strategies.

A. Operational requirements

We define a set of operational requirements to cover the potential smart road traffic monitoring application; These include:

- Real-time vehicle detection and counting: The sensor node needs to be able to detect and count vehicles. Edge application services reduce the volume of data that must be moved, the consequent traffic, and the distance that data must travel. This provides lower latency and reduces transmission costs.
- Low cost: It should be expected that the deployment will depend largely on low-cost IoT nodes resource-constrained in terms of memory, computing capabilities and energy consumption.
- Scalability: New nodes can be added at any time, meaning volunteers can easily deploy the node in their car/building and report the traffic status using edge computing based video analytics which in turn expand monitoring networks.

B. Proposed Architecture

Following the edge computing architecture and according to the above identified requirements, the system architecture we designed is shown in Figure 1. Edge computing is a distributed computing paradigm which brings computation and data storage closer to the location where it is needed,to improve response times and save bandwidth. It can be broadly viewed as the integration of 3 modules: (i) sensing and IoT (ii) edge (iii) cloud.

The sensing layer is the lowest layer of the architecture. At this layer, the device detects the vehicles when they pass by. In smart transportation, there are different parameters to be monitored including the number of vehicles on a given road segment at a given time interval. A low cost camera can be used as a sensor to capture video and perform video analytics to count the number of vehicles on the given road. In our proposed solution a camera sensor connected with the edge node performs real-time video processing for vehicle detection and classification. Only the processed and summarized data is sent from the edge node to the LoRaWAN gateway, which reduces bandwidth usage.

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A LoRa-enabled device is necessary to interface with the edge node and communicate the readings to a nearby gateway. This device needs to have sufficient memory capacity to store detection information while operating with minimal power consumption to ensure long lifetime. Gateways are vital components of a LoRaWAN network, responsible for the collection and forwarding of any frames they receive, handing them off to the back-end application server. They also provide other critical functionality including protocol translation, data filtering, security and management.

Finally, the third layer comprises a cloud server. It performs aggregated analysis on data received from edge nodes for further use in traffic analysis and decision making.

The flow chart shown in Figure 2 depicts the whole operation of the system. The Raspberry Pi Camera Module is responsible for acquiring the video and making frame by frame analysis for vehicle detection and counting.'It connects to the LoPy either using UART or by creating a wireless local area network using Bluetooth or WiFi. Once the LoPy receives the vehicle count, it sends it to the cloud over the LoRaWAN network.

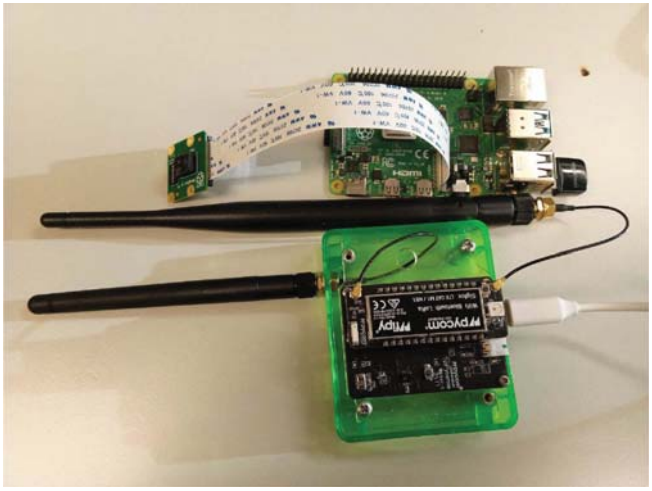


Fig. 3. The Raspberry Pi with the camera and a LoRa LoPy transceiver.

IV. Experiment and Evaluation

In this section we describe first our experiment setup and then its performance using different criteria.

To maximize the relevance and applicability of our model, we selected hardware devices that are affordable and widely adopted. Raspberry Pi is the most popular single board computer. LoPy from PyCom [27] were used for the LoRa transceiver as shown in Figure 3.

The original YoloV3 [6] was written with a C++ library called Darknet [28]. We deployed our own model, YoloV3-VTiny, with four classes (bicycle, car, motorbike, truck) on Raspberry Pi Model 3B and a Raspberry Pi Model 4B.

LoRaWAN architecture follows a star topology, in which gateways relay messages between end nodes and network servers. Gateways are connected to the network server using any kind of IP connection and act as a bidirectional bridge. In our work we used the widely available TTN platform (The Things Network) [29], but alternatives like ChirpStack [30], formerly known as LoRa server, can also be used.

The payload length has a relevant impact on the time on air of the packets. In our experiment, the payload has a length of 31 bytes comprising vehicle count, location and timestamp.

Data from the network server can be obtained using various APIs. We used MQTT (Message Queuing Telemetry Transport). The network server provides as metadata the various details of the data channel (application and device identifiers, frame count, frequency in MHz, airtime in ns, rssi in dBm and the gateways that received packet) as shown in Listing 1.

V. Results

In this section we evaluate the accuracy and performance of the system in detecting and counting the vehicles and its network performance.

A. Accuracy

We evaluated the vehicle counting models using offline videos and live cameras on the road side as well as on moving vehicles. Table 1 summarizes the performance results with basic statistics computed on offline video. To assess the accuracy percentage of the detection, we measure the number of vehicles detected by the sensor and compare it with the ground truth using the following formula:

$$Accuracy = (\frac{TN + TP}{TotalCount}) * 100 \quad (1)$$

where TP - True Positive means there are indeed cars on the monitoring area and they were detected correctly, TN - True Negative means there are no cars in the monitoring area and the vehicle detector didn't registered any, while Total Count corresponds to the total number of vehicles independently measured (ground truth). Results are given in Figure 6.

For live assessment, as shown in Figure 4, we installed the end node on the road side and then measured the traffic flow. The end node communicates with the gateway to report its reading to TTN. Table 2 summarizes the performance results with basic statistics computed on the live webcam.



Fig. 4. Live Vehicle Detection and Counting

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Listing 1. Example of packet received by the gateway


```
{
    "app_id": "vehicle_counting",
    "dev_id": "car1",
    "hardware_serial": "70B3D5499ABCE42A",
    "port": 2,
    "counter": 58,
    "payload_raw": "MzMPQT0KG0IAAMBAM...",
    "payload_fields": {
        "lat": 8.949999809265,
        "lng": 38.7599983215332,
        "vehicle_count": 6
    },
    "metadata": {
        "time": "2020-06-04T00:47:31.27Z",
        "frequency": 868.1,
        "modulation": "LORA",
        "data_rate": "SF7BW125",
        "airtime": 92416000,
        "coding_rate": "4/5",
        "gateways": [
            {
                "gtw_id": "eui-cc50e3...",
                "timestamp": 555258263,
                "time": "2020-06-04T00:...",
                "channel": 0,
                "rssi": -124,
                "snr": -7,
                "rf_chain": 0
            }
        ]
    }
}
```


TABLE I
PERFORMANCE RESULTS WITH BASIC STATISTICS COMPUTED ON OFFLINE VIDEO

Platform	Traffic Video	Model	Model loading (sec)	Avg. accuracy
Laptop	New delhi	YOLO3	0.798695	96%
Laptop	New delhi	YoloV3-VTiny	0.798695	93%
Laptop	Nigera	YOLO3	0.798695	97%
Laptop	Nigera	YoloV3-VTiny	0.0136	94%
Laptop	Addis Ababa	YOLO3	0.798695	98%
Laptop	Addis Ababa	YoloV3-VTiny	0.0136	94%
RP3	New delhi	YOLO3	0.9493	73%
RP3	New delhi	YoloV3-VTiny	0.1389	65%
RP3	Nigera	YOLO3	0.9347	74%
RP3	Nigera	YoloV3-VTiny	0.1364	66%
RP3	Addis Ababa	YOLO3	0.9385	76%
RP3	Addis Ababa	YoloV3-VTiny	0.1375	65%
RP4	New delhi	YOLO3	0.9493	79%
RP4	New delhi	YoloV3-VTiny	0.1389	69%
RP4	Nigera	YOLO3	0.9388	80%
RP4	Nigera	YoloV3-VTiny	0.1364	70%
RP4	Addis Ababa	YOLO3	0.9385	82%
RP4	Addis Ababa	YoloV3-VTiny	0.1375	71%

TABLE II
PERFORMANCE RESULTS WITH BASIC STATISTICS COMPUTED ON LIVE WEBCAM

Platform	Model	fps	Actual # of Vehicles	Counted # of Vehicles	Average accuracy
RP3	YOLO3	15.17	15	12	81%
RP3	YOLO-VTiny	14.00	15	11	73.33%
RP4	YOLO3	19.57	15	12	81%
RP4	YOLO-VTiny	18.54	15	11	73.33%

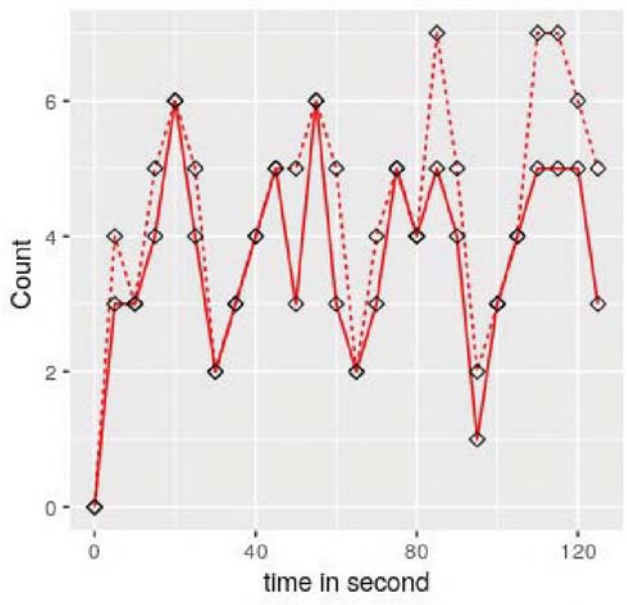


Fig. 5. Time Series comparison of vehicles count by the system (continuous line) vs. ground truth (dashed line).

B. LoRaWAN

We also evaluated the latency of the LoRaWAN network for the proposed system. The payload of the data transmitted was 31 bytes containing GPS location and count of vehicles over the previous 5 seconds. In the worst case scenario, with a LoRaWAN data rate configuration 0, (corresponding to a spreading factor of 12 and a bandwidth of 125 kHz in the EU868 band) the latency is 2138.1 ms. However, it is reduced to 71.936 milliseconds with a configuration data rate of 5 (Spreading factor 7, 125 kHz bandwidth) and coding rate of 4/5.

In any system there is always a trade off between resource usage and accuracy of the outputs. Choosing appropriate configurations in video analytics has considerable impact on the resource usage and accuracy of results.

We noticed a performance improvement by using multi thread, running camera I/O in a separate thread and feeding its output to the main process. These results have shown that a low-cost solution is feasible and effective. The system generated time-series data on road traffic can be analyzed promptly, a significant improvement on previous solutions, and economically transmitted over LoRaWAN for further processing.

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Traffic data can be analyzed to understand its behavior, identify any unexpected phenomena and even propose improvements in the road infrastructure. This can assist key stakeholders such as Municipal Traffic Managers, Operations Managers, and Town Planners. The testing and experimentation process has verified that the primary objectives laid out at the outset of the project have been fulfilled by quantifying a series of tangible results.

Despite concerns with duty cycle and payload limitations, LoRaWAN was able to provide long-range, near real-time traffic updates, without recurring operational costs. Using SF 12 we can send data every 3 minutes and with SF 7 every 1.5 minutes. This means that it can also be used in similar applications, especially in developing countries.

VI. Conclusion

This paper presented a low-cost, real time road traffic monitoring solution using IoT and machine learning for vehicle detection and counting to provide updated traffic information for better ITS. It shows the potential for low cost emerging technologies, including LoRaWAN, IoT devices, and advanced machine learning techniques to be adopted for a wide range of urban computing applications.

These solutions provide significant opportunities for cities in developing countries that encounter daily traffic jams and have poor road traffic management systems. The proposed detection system can benefit future cities through improved reliability, lower costs and improved scalability, presenting a positive impact on cities' sustainability.

We developed a prototype and measured its performance that shows its suitability. Future work include the improvement of the detection and classification under different weather conditions and time of the day, as well as assessing the performance when a significant number of sensors is deployed in the same area, which may raise concerns about the effects of interference and duty cycle limitations. Furthermore we will focus on the analytical side of the system, including time-series data analysis geared towards building predictive models for urban road traffic behavior.

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